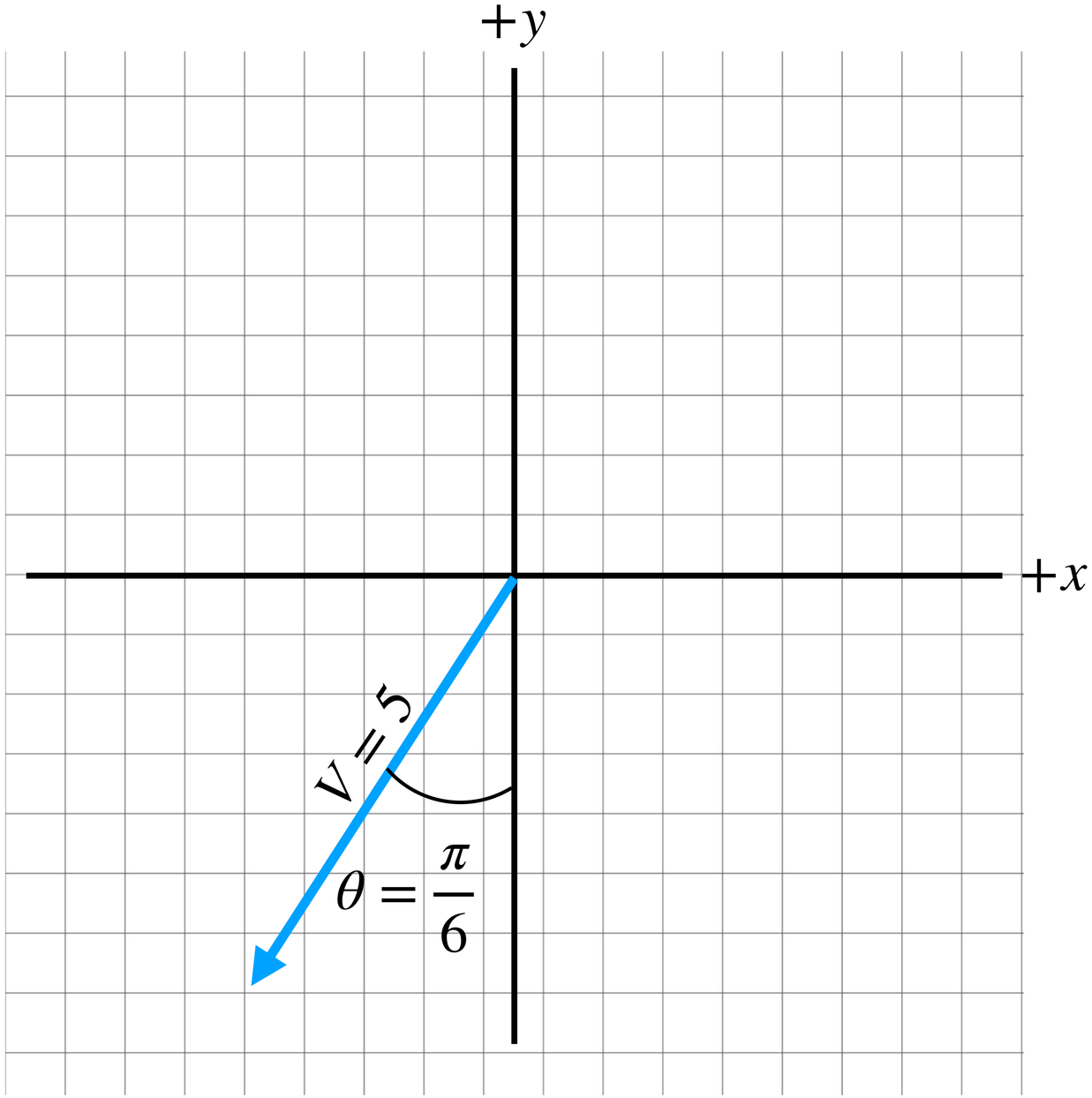
How are you feeling?

- Pacing of lectures?
- Clarity of material?



Check yourself

- What are the components of $\vec{V}$?



Check yourself

Given the following vectors

$$\overrightarrow{V}_1 = 2\hat{i} + 3.5\hat{j} - \hat{k}$$

$$\overrightarrow{V}_2 = -0.4\hat{i} + 4\hat{j} + 1.2\hat{k}$$

- What is $\overrightarrow{V}_1 + \overrightarrow{V}_2$?
- What is $\overrightarrow{V}_2 - \overrightarrow{V}_1$?
- What is $\frac{1}{2}\overrightarrow{V}_2$?

Motion

What is motion?

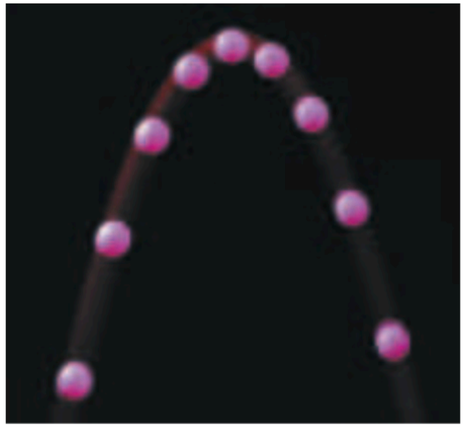
- Motion: the change of an object's position with time
- Four basic types of motion
- Object moves through space: **translational motion**
- The path an object moves along: **trajectory**



Linear motion



Circular motion



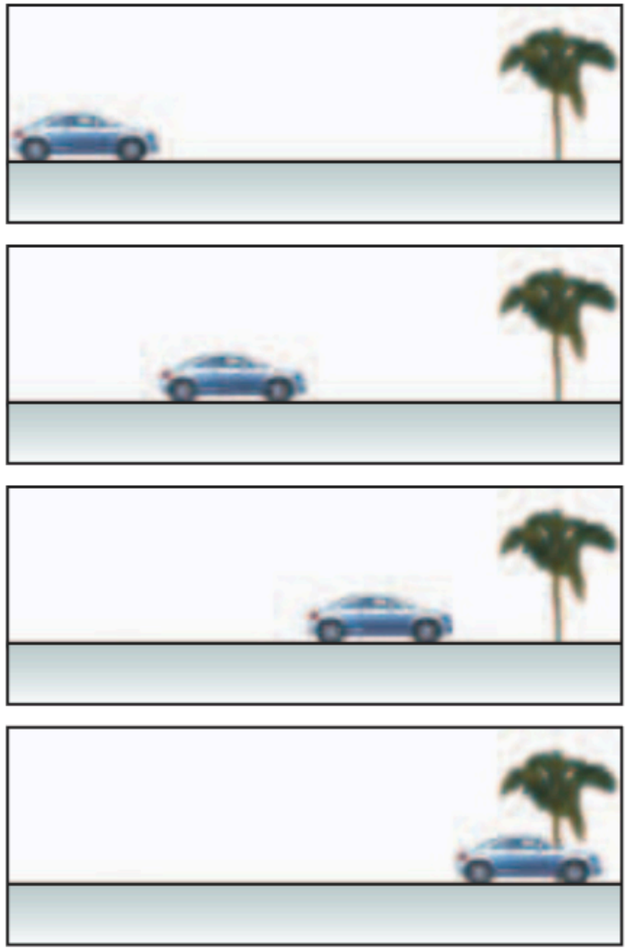
Projectile motion



Rotational motion

Motion Diagrams

- Tool for visualizing and building intuition motion
- Diagram of object's position at **equally spaced** instances of time
  - Like frames of video layered on top of each other

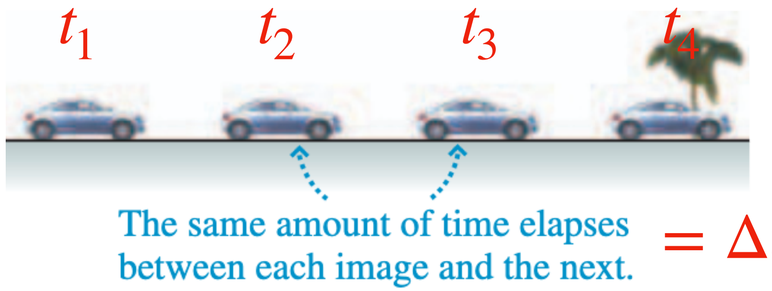


Frame 1: t_1

Frame 2: $t_2 = t_1 + \Delta t$

Frame 3: $t_3 = t_2 + \Delta t$

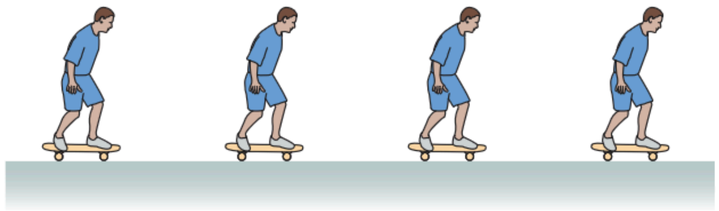
Frame 4: $t_4 = t_3 + \Delta t$

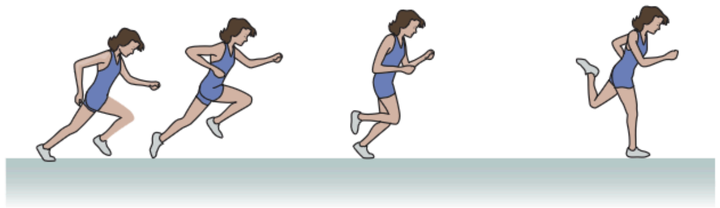


Δ = Greek letter delta (indicates change of quantity)

What do you think?

- In the following motion diagrams:
  - Is the time between each snapshot of object the same?
  - Is the space between each snapshot of object equally? If no, is the spacing increasing or decreasing?
  - Is the object speeding up, slowing down, or no change of speed?



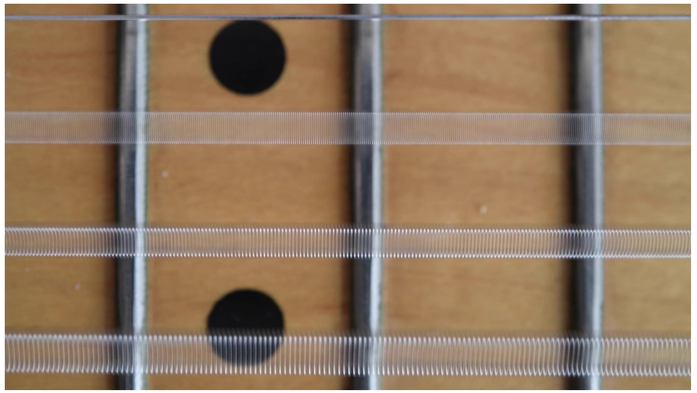




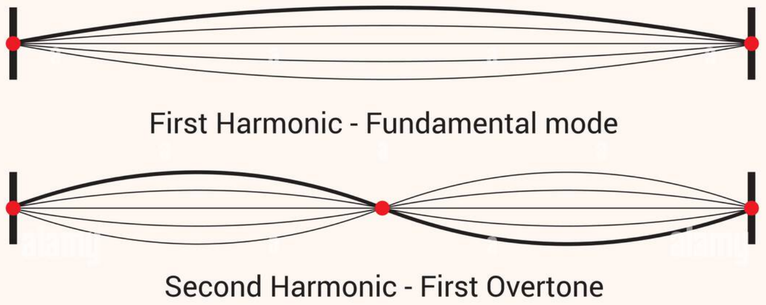
Models and modeling

- Real world complex
- Model: simplified picture of reality that captures essential features
- **Particle model**: object represented as a mass at a single point
  - Particle has **no size**, **no shape**, **no orientation** (top, bottom, front, back, etc.) **only position and mass**
  - Particle model of motion treats an object's motion as motion of particle

Real world



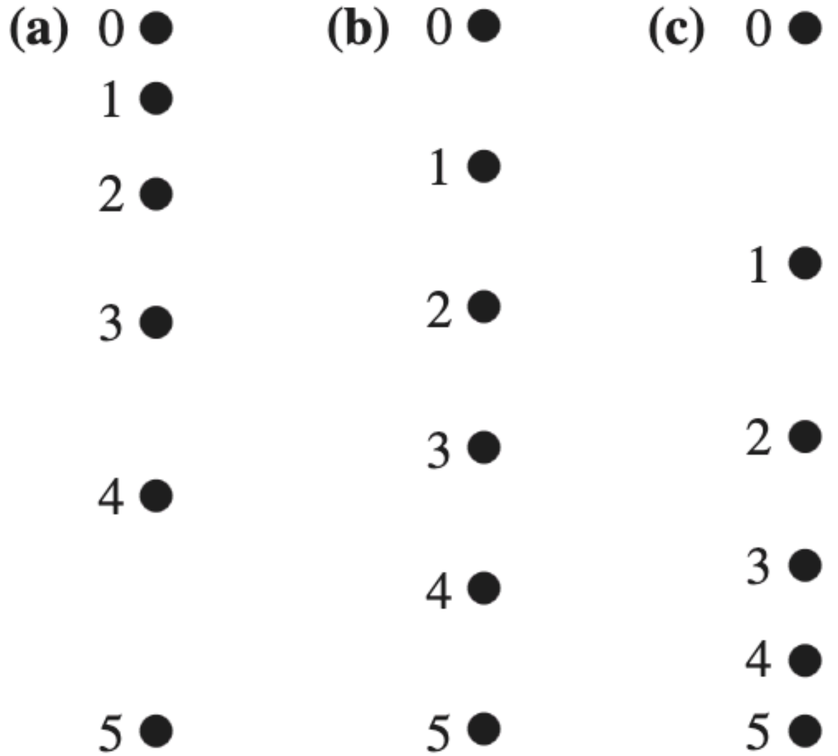
Model



What do you think?

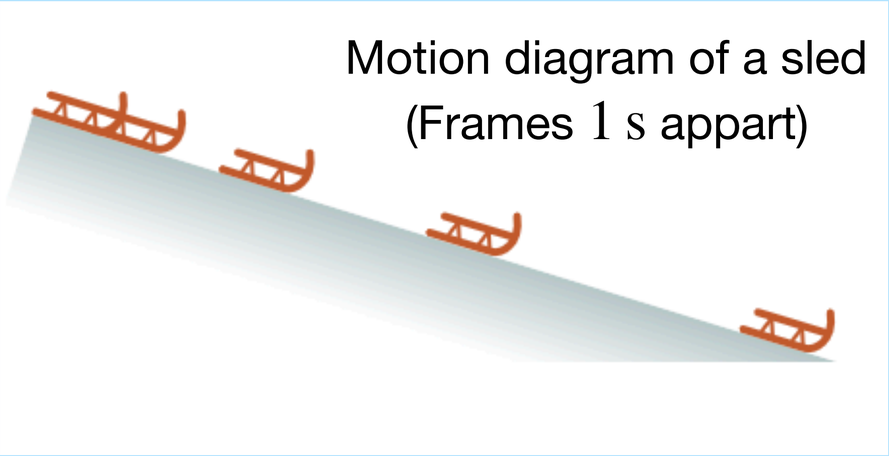
- Which of the motion diagrams is:

  1. A dust particle settling to the floor at constant speed
  2. A descending rocket slowing to make a soft landing on mars?
  3. A ball dropped from the roof of a building

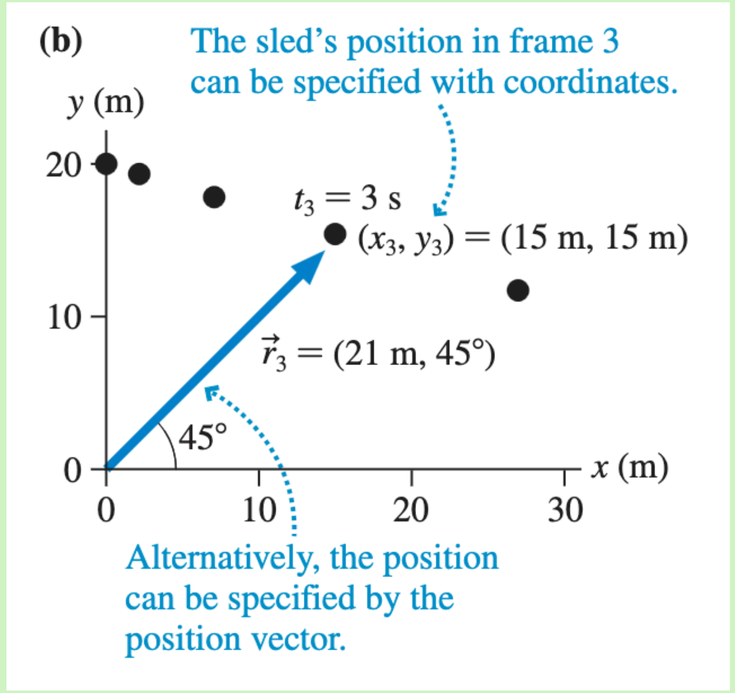


Position

- Where an object is in space (given a coordinate system)
- Vector that points from origin to object
  - Equal to object's coordinates
- Common to denote as $\vec{r}$

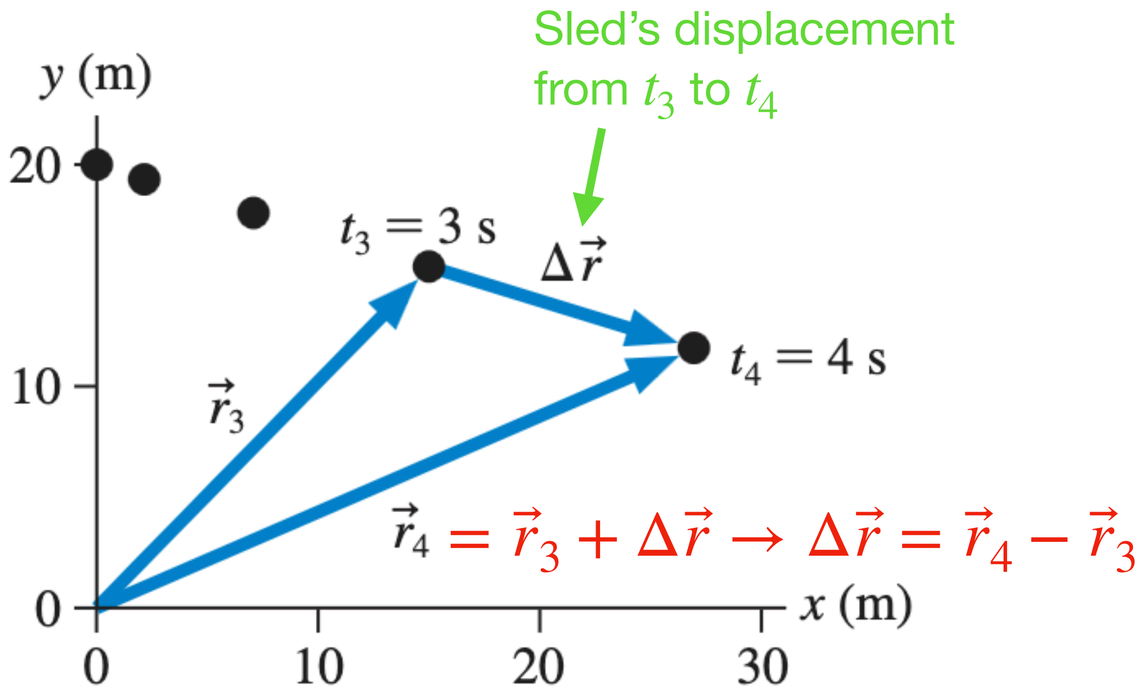


Motion diagram as particle and plotted



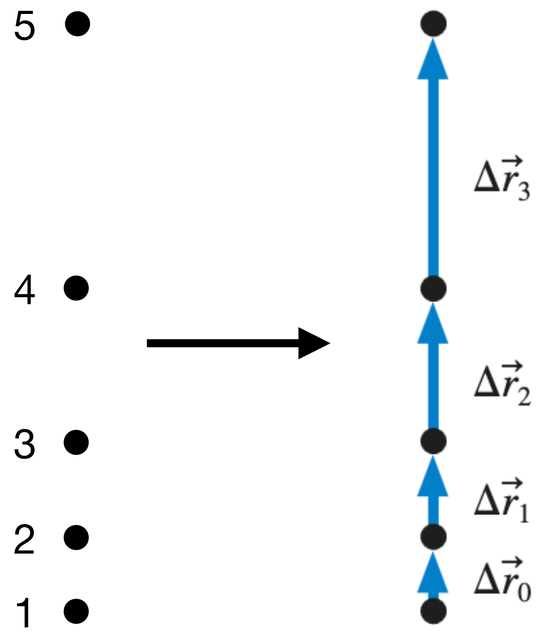
Displacement

- Change in an object's position with time
- Vector
- Common to denote as $\Delta\vec{r}$
- $\Delta\vec{r} = \vec{r}_b - \vec{r}_a$: displacement of object as it moves from $\vec{r}_a$ to $\vec{r}_b$



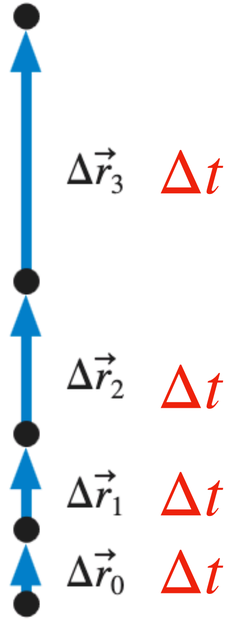
Motion diagrams with displacement vectors

- Displacement vectors = arrows connecting dots
- Label each vector as $\Delta\vec{r}_n$
- **Length** of displacement vectors **increases**: object **speeding up**
- **Length** of displacement vectors **decreases**: object **slowing down**



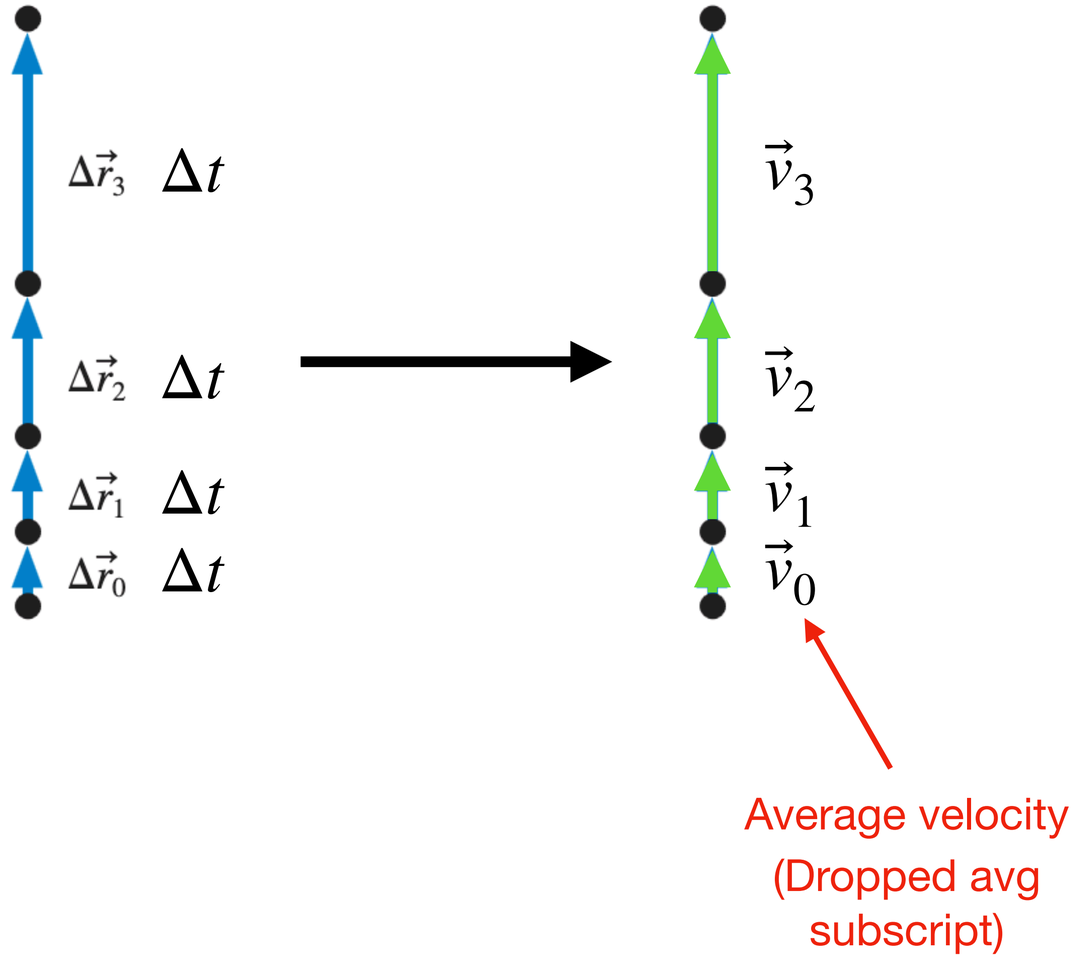
Time interval

- Elapsed time as object moves from position $\vec{r}_a$ at t_a to $\vec{r}_b$ at t_b
- $\Delta t = t_b - t_a$
- For a motion diagram Δt is constant



Velocity

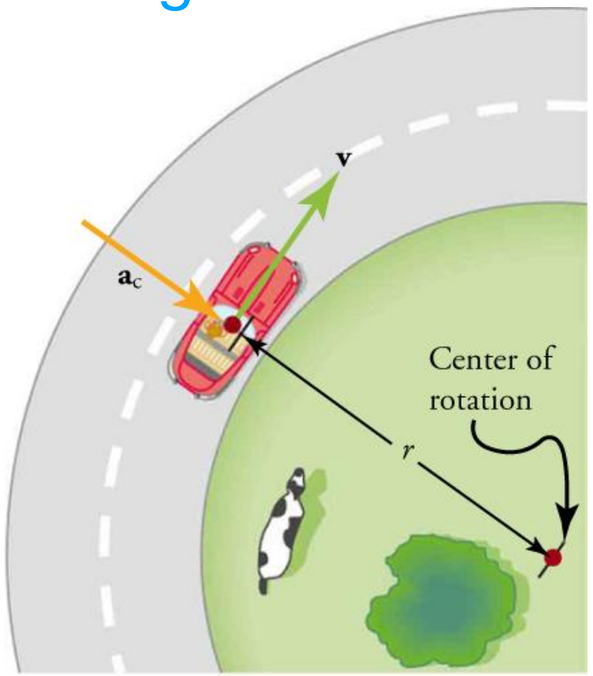
- Change in position over change in time
- Average velocity $\vec{v}_{\text{avg}}$
  - For given Δt and $\Delta \vec{r}$: $\vec{v}_{\text{avg}} = \dfrac{\Delta \vec{r}}{\Delta t}$
  - $\vec{v}_{\text{avg}}$ points in the same direction as $\Delta \vec{r}$
- Motion diagram: have $\Delta \vec{r}_n$ and Δt so each dot connected by average velocity vector



Acceleration

- Change in velocity over change in time
  - Magnitude (change in *speed*)
  - Direction (change in direction)
- Linear acceleration = change in magnitude **not** direction
- Average acceleration $\vec{a}_{\text{avg}}$
  - For given Δt and $\Delta \vec{v}$: $\vec{a}_{\text{avg}} = \dfrac{\Delta \vec{v}}{\Delta t}$
  - $\vec{a}_{\text{avg}}$ points in the same direction as $\Delta \vec{v}$

Change in direction



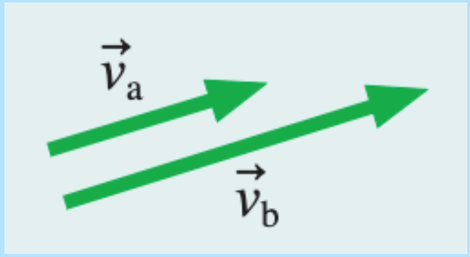
Change in speed



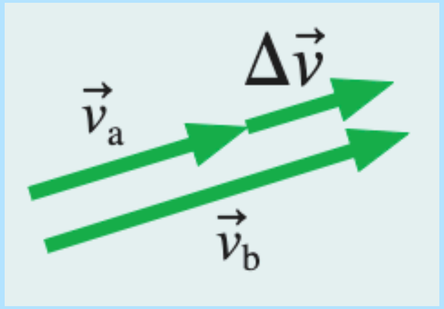
Finding acceleration vectors on motion diagram

Goal: determine change of velocity $\Delta\vec{v} = \vec{v}_b - \vec{v}_a$

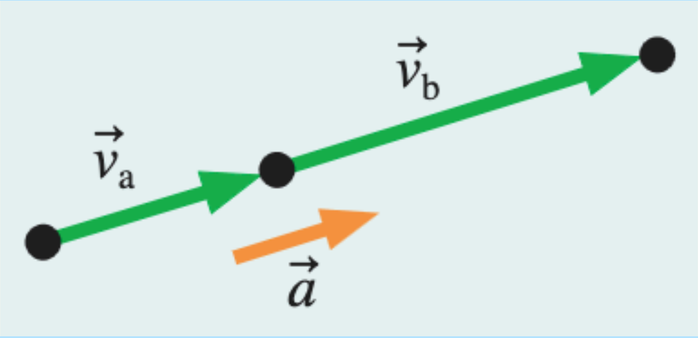
1. Draw velocity vectors $\vec{v}_a$ and $\vec{v}_b$ with their tails together



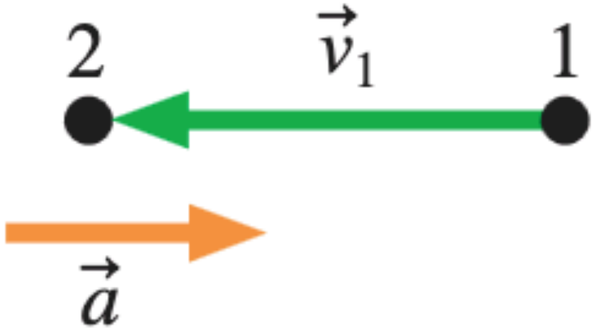
2. Draw the vector from the tip of $\vec{v}_a$ to the tip of $\vec{v}_b$ to get $\Delta\vec{v}$



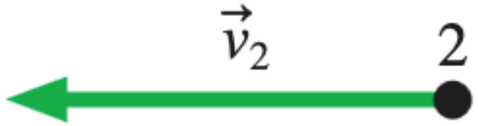
3. Draw $\Delta\vec{v}$ at dot between $\vec{v}_a$ and $\vec{v}_b$; label $\vec{a}$



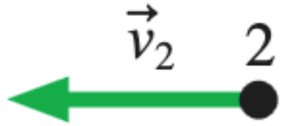
What do you think?



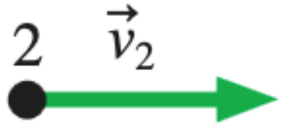
In the figure above, a particle undergoes acceleration $\vec{a}$ while moving from point 1 to point 2. Which of the choices shows the most likely velocity vector $\vec{v}_2$ as the particle leaves point 2?



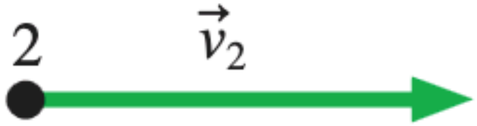
(a)



(b)



(c)



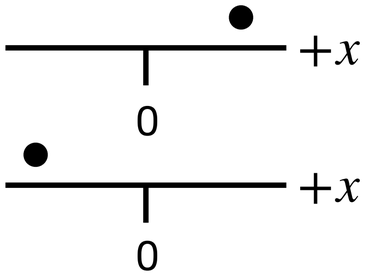
(d)

Introduction to 1D Motion

1D Motion

- Position $\vec{r}$, velocity $\vec{v}$, and acceleration $\vec{a}$ vectors point along one axial direction
- Simplified notation, e.g., (x, v_x, a_x) or (y, v_y, a_y)
- The sign (if a vector is positive or negative) carries all information about direction

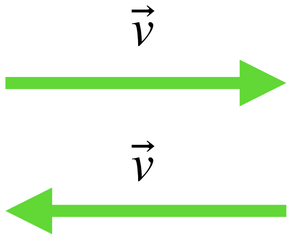
Sign vectors and direction



$x > 0$: Position to right of origin

$x < 0$: Position to right of origin

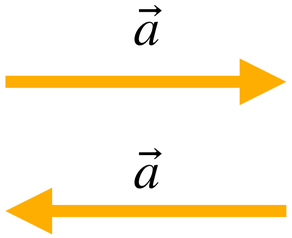
Sign tells us *where* an object is



$v_x > 0$: Direction of motion to the right

$v_x < 0$: Direction of motion to the left

Sign tells us *which direction* the object is moving



$a_x > 0$: Acceleration points to the right

$a_x < 0$: Acceleration points to the left

Sign tells us which way the acceleration points (**not** if the object is speeding up or down)

For 1D motion

- An object is speeding up **if and only if** v_x and a_x have the same sign
- An object is slowing down **if and only if** v_x and a_x have opposite signs
- An object's velocity is constant **if and only if** $a_x = 0$

Position vs time graphs

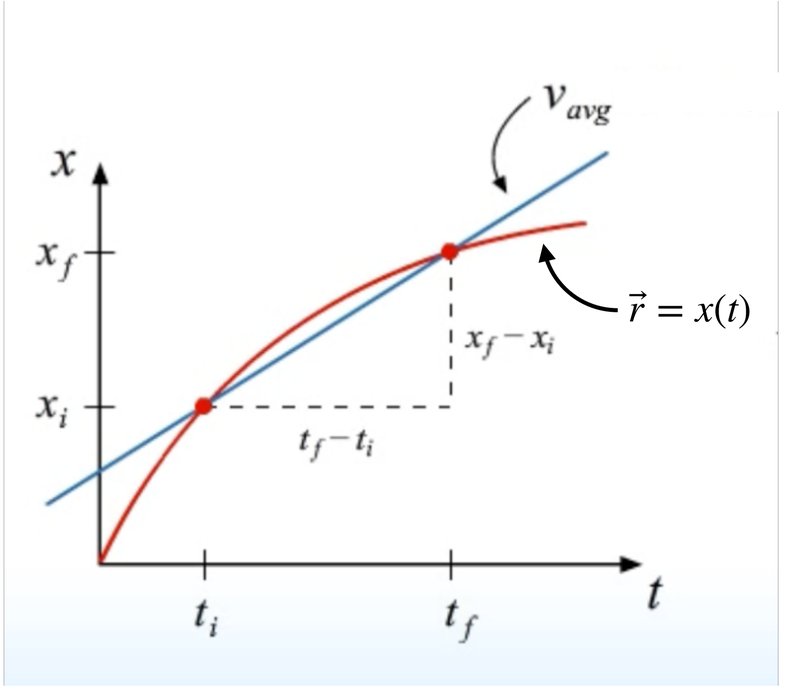
- Another representation of motion
- Position as a function of time (i.e., position on vertical axis, time on horizontal)
- Slope at any time t is the velocity

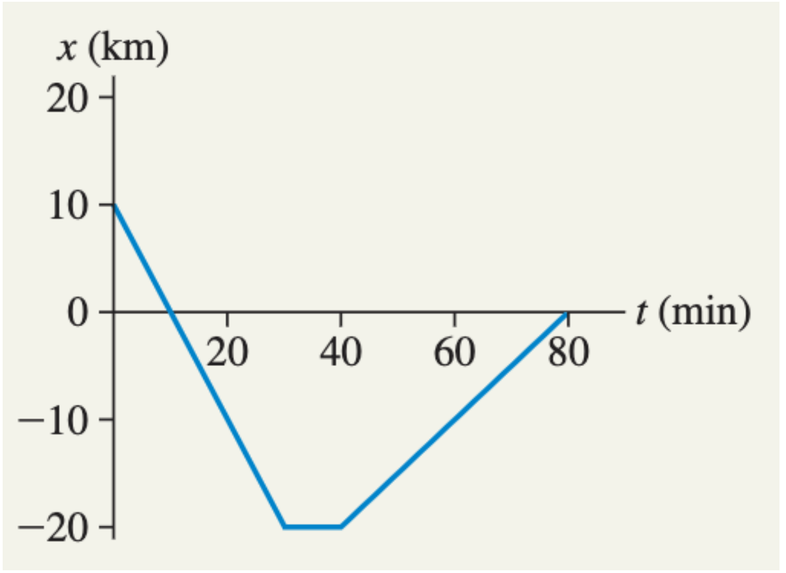
$$v_x = \frac{\mathrm{d}x}{\mathrm{d}t}$$

- Slope between two times is average velocity

$$v_{x,\mathrm{avg}} = \frac{x_\mathrm{f} - x_\mathrm{i}}{t_\mathrm{f} - t_\mathrm{i}}$$

- Steeper slopes = faster speed
- Negative slopes = negative velocity = motion in the negative direction





Velocity graph

- Any instant of time = slope of position graph
- Slope is acceleration

$$a_x = \frac{dv_x}{dt}$$

- Slope between two times is average acceleration

$$a_{x,\text{avg}} = \frac{v_f - v_i}{t_f - t_i}$$

- Constant slope of position graph = constant velocity in velocity graph

Position graph Velocity graph

